

Course Code: B23IT3101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ADVANCED JAVA					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the syntax to establish a database connection using Driver Manager in JDBC.	1	2	2
	b).	Define the purpose of the ResultSet interface in JDBC	1	2	2
	c).	List any two differences between a Web Server and an Application Server.	2	2	2
	d).	Explain the role of a web container in a J2EE application	2	2	2
	e).	What is the role of ServletContext in a web application?	3	2	2
	f).	What is the significance of the service() method in the Servlet Lifecycle?	3	2	2
	g).	Write any two implicit objects used in JSP and their purposes.	4	2	2
	h).	Explain the purpose of the page directive in JSP	4	2	2
	i).	List any two annotations used in Spring MVC and mention their usage.	5	2	2
	j).	Describe the role of the @Autowired annotation in Spring	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Describe how batch updates in JDBC improve performance compared to individual updates, and provide an example scenario where batch updates would be beneficial.	1	3	7
	b).	Explain the role of the JDBC architecture	1	2	3
		OR			
3.		Create a JDBC application to perform CRUD operations (Create, Read, Update, Delete) on a "Courses" table with columns course_id, course_name, and credits.	1	3	10
		UNIT-2			
4.	a).	Design a basic directory structure of a web application using J2EE.	2	2	6
	b).	What are the key differences between HTTP GET and POST methods? Explain how these methods are used in a J2EE web	2	2	4

		application.			
		OR			
5.		Given a typical web application request (e.g., submitting a user registration form), trace the sequential steps involved in its processing by a web server and a web container.	2	3	10
		UNIT-3			
6.	a).	Explain the Servlet Life Cycle and the purpose of the init, service, and destroy methods in a servlet.	3	2	5
	b).	Describe how session tracking is achieved in servlets using cookies and HTTP session, and compare their advantages and limitations.	3	2	5
		OR			
7.		Write a servlet that uses RequestDispatcher to forward a user's login request to a welcome page if credentials are valid, or redirects to an error page using send Redirect if invalid.	3	3	10
		UNIT-4			
8.	a).	Explain the differences between JSP and Servlets, and describe the role of JSP directives in controlling page behaviour.	4	2	6
	b).	Describe the purpose of Expression Language (EL) in JSP and how implicit objects (e.g., request, session) are accessed using EL.	4	2	4
		OR			
9.		Write a JSP page that uses JSTL to display a table of products from a List stored in the session, including columns for product_id, name, and price.	4	3	10
		UNIT-5			
10	a).	Describe how dependency injection is implemented in Spring using annotations like @Autowired, and explain the differences between constructor injection and setter injection.	5	2	7
	b).	Describe the role and significance of a "bean" in the context of the Spring Framework.	5	2	3
		OR			
11	a).	What is the primary benefit of utilizing the Spring DAO module for database operations?	5	2	5
	b).	Describe how Spring's transaction management ensures data consistency in database operations.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER NETWORKS					
For IT					
Time: 3 Hrs.		Max. Marks: 70 M			
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define a) Simple Analog Signal b) Composite analog signal	1	1	2
	b).	What do you mean by a) Layers b) Protocol	1	1	2
	c).	Explain the characteristics of twisted pair cable.	2	2	2
	d).	Calculate 4-bit checksum for the message 1010110101010100	2	2	2
	e).	What is the significance of variable length framing	3	2	2
	f).	Differences between FDMA and TDMA	3	2	2
	g).	What is CIDR addressing?	4	1	2
	h).	What is the purpose of TTL field in IPV4	4	1	2
	i).	Write uses of User Datagram Protocol(UDP)	5	2	2
	j).	What are the two main categories of DNS messages?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain briefly about the TCP/IP Model.	1	2	5
	b).	Different types of networks	1	1	5
		OR			
3.	a).	Explain briefly about the ISO-OSI Model.	1	2	5
	b).	Discuss about Analog signals and digital signals	1	2	5
		UNIT-2			
4.	a).	Given 1101011011 data frame and generator polynomial $G(x) = x^4 + x + 1$. Derive the transmitted frame	2	2	5
	b).	What are the Types of Wireless Transmission Media? Explain it.	2	1	5
		OR			
5.	a).	Discuss a) Packet Switching b) Circuit Switching	2	1	5
	b).	Discuss about error detection and correction code with example	2	2	5
		UNIT-3			

6.	a).	Explain CSMA/CD protocol and how it detects collision?	3	2	5
	b).	Compare various sliding window protocols of data link layer	3	2	5
		OR			
7.	a).	How performance is improved in CSMA/CD protocol compared to CSMA protocol? Explain?	3	2	5
	b).	Explain modes and frames of HDLC	3	2	5
		UNIT-4			
8.	a).	Consider an address block 121.37.10.64 /26. Find the first and last addresses for each subnet, if 4 no of subnets required is as given in the input.	4	2	5
	b).	Describe the problem and solutions associated with distance vector routing.	4	2	5
		OR			
9.	a).	Given network address of 192.18.100.0 and a subnet mask of 255.255.255.192 . a) How many subnets are created? b) How many hosts are there per subnet?	4	2	5
	b).	Discuss ICMP Messages.	4	1	5
		UNIT-5			
10.	a).	Explain about Application layer and its services in detail?	5	2	5
	b).	Describe TCP connection management. 5M	5	2	5
		OR			
11.	a).	Discuss the features of HTTP and discuss how HTTP works	5	2	5
	b).	Describe Avoidance of congestion in TCP	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

III B.Tech. I Semester MODEL QUESTION PAPER

AUTOMATA THEORY & COMPILER DESIGN

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

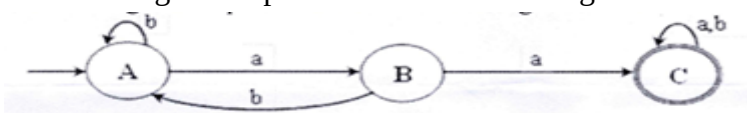
Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Explain the Chomsky Hierarchy of languages.	1	2	2
	b).	Explain the Kleen and Positive closures of a language.	1	2	2
	c).	List the applications of Pumping Lemma.	2	2	2
	d).	Write algebraic laws for Regular Expressions	2	2	2
	e).	Mathematical Representation of Push Down Automata.	3	2	2
	f).	Define Non deterministic Turning Machine	3	2	2
	g).	State Token, Lexeme, Pattern.	4	2	2
	h).	Explain the Top Down and Bottom-Up Parsing.	4	2	2
	i).	Define Syntax Directed Definition	5	2	2
	j).	List the Issues in the Design of Code Generator	5	2	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M															
2.	a).	Design DFA to accept set of binary strings of 0's and 1's only those that are divisible by 3.	1	3	5															
	b).	Convert the following NFA to DFA <table><tr><td></td><td>0</td><td>1</td></tr><tr><td>→p</td><td>{p,q}</td><td>{p}</td></tr><tr><td>q</td><td>{r}</td><td>{r}</td></tr><tr><td>r</td><td>{s}</td><td>∅</td></tr><tr><td>*s</td><td>{s}</td><td>{s}</td></tr></table>		0	1	→p	{p,q}	{p}	q	{r}	{r}	r	{s}	∅	*s	{s}	{s}	1	3	5
	0	1																		
→p	{p,q}	{p}																		
q	{r}	{r}																		
r	{s}	∅																		
*s	{s}	{s}																		
		OR																		
3.	a).	Design a DFA to accept set of binary strings of 0's and 1's which do not contain substring 01.	1	3	7															
	b).	List and explain the applications of finite automata.	1	2	3															
		UNIT-2																		

4.	a).	Find the Regular Expression for the following finite Automaton. 	2	3	5
	b).	Show that the grammar is ambiguous $S \rightarrow a \mid abSb \mid aAb$ $A \rightarrow bs \mid aAAb$	2	3	5
		OR			
5.	a).	Obtain finite automata for regular expression denoted by $(0^*+1^*+2^*)^*$.	2	3	5
	b).	Define CFG. Consider the following CFG $S \rightarrow 0B \mid 1A$ $A \rightarrow 0 \mid 0S \mid 1AA$ $B \rightarrow 1 \mid 1S \mid 0BB$ Derive LMD and RMD for the string 00110101	2	3	5
		UNIT-3			
6.	a).	Explain the types of CFL Acceptance in PDA?	3	2	3
	b).	Design a Turing Machine to recognize the Language $L = \{0^n 1^n \mid n \geq 1\}$	3	3	7
		OR			
7.	a).	Design a PDA for the following CFG $S \rightarrow 0A$ $A \rightarrow 0AB \mid 1$ $B \rightarrow 1$	3	3	5
	b).	Differentiate between Turning Machine and Push Down Automata.	3	3	5
		UNIT-4			
8.	a).	Explain Phases of compiler with a neat Sketch.	4	2	5
	b).	Construct the LL(1) parsing table for the following grammar. $E \rightarrow T+E \mid T$ $T \rightarrow \text{float} \mid \text{float} * T \mid (E)$	4	3	5
		OR			
9.	a).	Elaborate specification and recognition of tokens	4	2	5
	b).	Construct LALR(1) parser for the following grammar. $S \rightarrow L=R \mid R$ $L \rightarrow *R \mid \text{id}$ $R \rightarrow L$	4	3	5
		UNIT-5			
10.	a).	Explain Peephole Optimization.	5	2	5
	b).	Design DAG for the following basic block: $D := B - C$ $E := A + B$ $B := B + C$ $A := E - C$	5	3	5
		OR			
11.	a).	Write the quadruples, triples and indirect triples for the expression: $(a+b)*(c+d)*(a+b+c)$	5	2	5
	b).	Explain the simple code generation algorithm with example.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

III B.Tech. I Semester MODEL QUESTION PAPER

OBJECT ORIENTED ANALYSIS AND DESIGN

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define object model.	1	1	2
	b).	Explain why an object-oriented approach is preferable when compared to other approaches?	1	2	2
	c).	What is key abstraction?	2	1	2
	d).	List the relationships among classes.	2	1	2
	e).	What is the importance of modeling?	3	1	2
	f).	Explain annotational things.	3	2	2
	g).	What is an Usecase?	4	1	2
	h).	Define include and extend relationships used in usecase diagram.	4	1	2
	i).	What is the difference between node and device.	5	1	2
	j).	Define component diagram.	5	1	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M
2.	a).	Explain the structure of complex systems.	1	2	5
	b).	Interpret the evolution of object model.	1	2	5
		OR			
3.	a).	Interpret the attributes of complex systems.	1	2	5
	b).	Explain Organized and Disorganized Complexity.	1	2	5
		UNIT-2			
4.	a).	Illustrate the relationship among Classes	2	2	5
	b).	How to identify Classes and Objects? Give an example	2	3	5
		OR			
5.	a).	Define an object. Illustrate common uses of objects with an example	2	2	5
	b).	Explain key abstractions and mechanisms	2	2	5
		UNIT-3			

6.	a).	Demonstrate the basic blocks of the UML	3	3	5
	b).	Design class diagram for “online job portal”.	3	3	5
		OR			
7.	a).	Design object diagram for “online health care system”.	3	3	5
	b).	Differentiate class diagram and object diagram with an example.	3	3	5
		UNIT-4			
8.	a).	Illustrate usecase diagram and its relationships with an example.	4	2	5
	b).	Compose the state chart diagram for “online pharmacy management system”.	4	3	5
		OR			
9.	a).	Define an activity diagram. And explain its internal behavior with an example.	4	2	5
	b).	Discuss Interaction diagrams with an example	4	2	5
		UNIT-5			
10.	a).	What are components? Show the stereotypes that apply to components	5	2	5
	b).	Describe deployment diagram. When will it be used in software life cycle	5	3	5
		OR			
11.	a).	Compose the component diagram for “unified library application”.	5	3	5
	b).	Differentiate the component diagram and deployment diagram.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
CYBER SECURITY					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	State any two examples for internal and external threats.	1	1	2
	b).	Summarize common HTTP exploits.	1	2	2
	c).	Identify four measures to ensure availability.	2	2	2
	d).	Name the three states of data with example for each.	2	2	2
	e).	Identify four examples of logical access controls.	3	2	2
	f).	Give an example for extended ACL.	3	2	2
	g).	Name any four threats to cloud computing.	4	1	2
	h).	Name any four security policies.	4	1	2
	i).	Define penetration testing.	5	1	2
	j).	Identify the four risk mitigation strategies.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Discuss TCP and UDP Vulnerabilities and attacks.	1	2	5
	b).	Explain SQL injections with examples.	1	3	5
		OR			
3.	a).	Discuss various attacks targeting the IP vulnerabilities.	1	2	5
	b).	Summarize how can you mitigate common network attacks.	1	3	5
		UNIT-2			
4.	a).	Differentiate host-based and network-based protection.	2	3	5
	b).	Explain the Cyber Security cube.	2	2	5
		OR			
5.	a).	Explain the BYOD policy.	2	2	5
	b).	Suggest few physical security methods to be taken in an organization.	2	3	5
		UNIT-3			
6.	a).	Summarize the threats on Embedded and IoT devices.	3	3	5

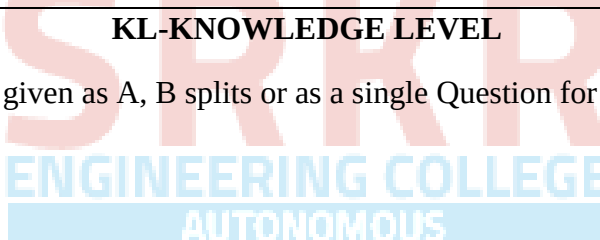
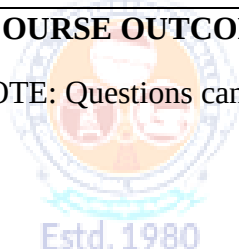
	b).	List and discuss various access control models.	3	2	5
		OR			
7.	a).	List and summarize various types of firewalls.	3	3	5
	b).	Explain ZPF.	3	2	5
		UNIT-4			
8.	a).	Discuss about the top threats on cloud computing.	4	3	5
	b).	Explain the types of security data.	4	2	5
		OR			
9.	a).	Identify any five types of security tests and give one example for each.	4	3	5
	b).	Summarize the phases of penetration testing.	4	2	5
		UNIT-5			
10.	a).	Discuss with examples sources of threat intelligence.	5	2	5
	b).	Explain Common Vulnerability Scoring System (CVSS).	5	2	5
		OR			
11.	a).	Explain weaponization and delivery steps in the cyber kill chain.	5	2	5
	b).	Explain the diamond model of intrusion analysis.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



Course Code: B23IT3106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ARTIFICIAL INTELLIGENCE					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Name two significant milestones in the history of AI.	1	1	2
	b).	What is the role of AI in autonomous vehicles?	1	2	2
	c).	What is backtracking in constraint satisfaction?	2	1	2
	d).	Define alpha and beta in the context of alpha-beta pruning.	2	1	2
	e).	What is a semantic tableau in propositional logic?	3	1	2
	f).	Explain resolution refutation in propositional logic.	3	2	2
	g).	Define extended semantic networks for knowledge	4	1	2
	h).	Define conditional probability, write its equation.	4	1	2
	i).	Name any three typical Expert Systems and their applications.	5	1	2
	j).	Define Hierarchical Planning.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Describe the evolution of AI from its inception to the present day.	1	2	5
	b).	What are intelligent agents? Describe different types of intelligent agents with examples.	1	2	5
		OR			
3.	a).	Describe the problem formulation and solution approach for Tic- Tac-Toe	1	2	5
	b).	Discuss the applications of AI in healthcare, education, and finance.	1	2	5
		UNIT-2			
4.	a).	Describe the iterative deepening A* algorithm and its advantages over A*.	2	2	5
	b).	Apply the principles of Constraint Satisfaction Problems (CSP) to solve the equation BASE+BALL=GAMES. Demonstrate the steps involved in finding the solution	2	3	5
		OR			
5.	a).	What is general problem solving? Explain its components with a	2	2	5

		suitable example			
	b).	Justify the use of Alpha-Beta Pruning in optimizing the Minimax Algorithm for game playing.	2	2	5
		UNIT-3			
6.	a).	Describe the natural deduction system in propositional logic. Explain the rules of inference with examples.	3	3	5
	b).	Use resolution refutation to prove the following statement is unsatisfiable: $(P \vee Q), (P \vee R), (Q \vee R)$	3	3	5
		OR			
7.	a).	Describe fuzzy logic and its components, including fuzzy sets, membership functions, and linguistic variables.	3	3	5
	b).	Prove the validity of the following argument using natural deduction: Premises - Premises: $P \Rightarrow Q \Rightarrow R, P$ - Conclusion: R	3	3	5
		UNIT-4			
8.	a).	Describe different approaches to Knowledge Representation, including Semantic Networks, Frames, and Logic-based Representation. Compare their advantages and disadvantages.	4	2	5
	b).	Write about Conceptual Dependency theory. How will it be used for Knowledge Representation?	4	3	5
		OR			
9.	a).	Represent the following knowledge using Frames: "John is a student. He studies Computer Science. He is 20 years old and lives in Hyderabad."	4	2	5
	b).	With an example, Explain Dempster-Shafer Theory in detail	4	2	5
		UNIT-5			
10.	a).	Describe the main components of any Planning System with a neat diagram.	5	2	5
	b).	Explain Reactive Systems in AI. How are they different from Deliberative Planning Systems?	5	3	5
		OR			
11.	a).	Explain the different phases involved in building an Expert system.	5	3	5
	b).	Illuminate the concept of Planning in Artificial Intelligence. Why is it important in problem-solving?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
MICROPROCESSORS & MICROCONTROLLERS					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List any four features of the 8085 microprocessors.	1	1	2
	b).	What is the function of the accumulator in 8085?	1	2	2
	c).	Define stack and its use in 8085.	2	1	2
	d).	What are the different types of interrupts in 8085?	2	1	2
	e).	Define memory-mapped I/O.	3	1	2
	f).	What is the role of ALE and IO/M signals in 8085?	3	2	2
	g).	What are the modes of operation of 8255?	4	1	2
	h).	Write the control word format of 8253.	4	1	2
	i).	List any two differences between microprocessor and microcontroller.	5	1	2
	j).	List the features of 8051 microcontroller	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the internal architecture of 8085 microprocessor with a neat block diagram.	1	2	5
	b).	Describe various addressing modes used in 8085 with suitable examples.	1	3	5
		OR			
3.	a).	Explain the timing diagram of STA 2050H instruction.	1	3	5
	b).	Describe the function of stack and subroutine with an example.	1	2	5
		UNIT-2			
4.	a).	Describe the procedure to interface an 8K RAM with 8085.	2	3	5
	b).	Discuss the importance and methods of address decoding.	2	2	5
		OR			
5.	a).	Explain the memory interfacing concepts with 8085 using 2732 EPROM and R/W memory.	2	3	5
	b).	Differences between memory mapped IO and peripheral IO	2	2	5

		UNIT-3			
6.	a).	Explain the block diagram and control word format of 8255 Programmable Peripheral Interface.	3	2	5
	b).	Describe the interrupt handling mechanism of 8259A with a block diagram.	3	2	5
		OR			
7.	a).	Discuss the block diagram and mode selection of 8253/54 Timer.	3	2	5
	b).	Explain the interfacing 8259 with the 8085 microprocessors.	3	2	5
		UNIT-4			
8.	a).	Explain the pin functions of 8086 in minimum mode.	4	2	5
	b).	Write short notes segmented memory in 8086?	4	2	5
		OR			
9.	a).	Explain the internal architecture of 8086 microprocessor with a neat block diagram.	4	2	5
	b).	Compare maximum and minimum mode of 8086 operation.	4	3	5
		UNIT-5			
10.	a).	Explain the Programming model of 8051 with all the components	5	2	5
	b).	Explain the internal RAM and special function registers of 8051.	5	2	5
		OR			
11.	a).	Draw the internal architecture of 8051 microcontrollers".	5	2	5
	b).	Explain the internal ROM, and special function registers of 8051.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
DATA WAREHOUSING & DATA MINING					
For IT					
Time: 3 Hrs.		Max. Marks: 70 M			
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List out various major issues to consider in the task of data mining.	1	1	2
	b).	List out the steps involved in the process of knowledge discovery.	1	1	2
	c).	Apply min-max normalization and map age value of 35 in the range [25,75] to the range [0.0,1.0]	2	3	2
	d).	Apply cosine similarity to find the similarity between following documents represented by term-frequency vector: Doc1(5,3,0,2,1,2) Doc2(4,2,2,2,2,3)	2	3	2
	e).	What is the problem of classification?	3	2	2
	f).	What is root node, internal node, and leaf node in a decision tree?	3	2	2
	g).	Explain support and confidence of an association rule: X→Y	4	2	2
	h).	What is meant by a closed frequent item set?	4	2	2
	i).	List out the major approaches to clustering.	5	1	2
	j).	Mention two limitations of k-means clustering.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Differentiate between OLTP systems and OLAP systems in various aspects.	1	2	5
	b).	Draw three-tier architecture for a data warehouse and explain the components of each layer.	1	2	5
		OR			
3.	a).	Explain the three schemas for multidimensional data models.	1	2	5
	b).	Explain briefly any three OLAP operations considering example of any three-dimensional data.	1	2	5
		UNIT-2			
4.	a).	Apply distance measures Euclidean and Manhattan to find the distance between objects represented by the tuples (22,1,42,10) and (20,0,36,8).	2	3	5

	b).	Explain about various types of attributes with suitable examples.	2	2	5														
		OR																	
5.	a).	Apply covariance and comment on the correlation of numeric attributes A and B: <table border="1"><tr><td>A</td><td>6</td><td>8</td><td>11</td><td>11</td><td>14</td></tr><tr><td>B</td><td>7</td><td>8</td><td>12</td><td>14</td><td>14</td></tr></table>	A	6	8	11	11	14	B	7	8	12	14	14	2	3	5		
A	6	8	11	11	14														
B	7	8	12	14	14														
	b).	Explain various sampling strategies for data reduction. Mention merits and demerits of each strategy.	2	2	5														
		UNIT-3																	
6.	a).	Explain clearly the process of inducing a decision tree.	3	2	7														
	b).	Explain briefly the bootstrap method used for evaluating the performance of a classifier.	3	2	3														
		OR																	
7.	a).	Explain the approach to classify a tuple using the naïve Bayes approach.	3	2	7														
	b).	Explain briefly rule-based and class-based ordering of rules generated by a rule-based classifier.	3	2	3														
		UNIT-4																	
8.	a).	Apply Apriori algorithm on the following set of transactions and determine all frequent item sets. Consider the minimum support count required as 4. <table border="1"><tr><td>TID</td><td>Items_Bought</td></tr><tr><td>T100</td><td>{I1,I2,I3,I4,I5}</td></tr><tr><td>T200</td><td>{I3,I4,I5,I6}</td></tr><tr><td>T300</td><td>{I1,I4,I5}</td></tr><tr><td>T400</td><td>{I1,I2,I4,I5}</td></tr><tr><td>T500</td><td>{I2,I4,I5,I6}</td></tr><tr><td>T600</td><td>{I1,I3,I4,I5}</td></tr></table> Clearly indicate the pruning step wherever applied.	TID	Items_Bought	T100	{I1,I2,I3,I4,I5}	T200	{I3,I4,I5,I6}	T300	{I1,I4,I5}	T400	{I1,I2,I4,I5}	T500	{I2,I4,I5,I6}	T600	{I1,I3,I4,I5}	4	3	7
TID	Items_Bought																		
T100	{I1,I2,I3,I4,I5}																		
T200	{I3,I4,I5,I6}																		
T300	{I1,I4,I5}																		
T400	{I1,I2,I4,I5}																		
T500	{I2,I4,I5,I6}																		
T600	{I1,I3,I4,I5}																		
	b).	Describe the methods to improve the efficiency of Apriori algorithm.	4	2	3														
		OR																	
9.	a).	Apply FP Growth algorithm on the following set of transactions and determine all frequent item sets. Consider the minimum support count required as 3. <table border="1"><tr><td>TID</td><td>Items_Bought</td></tr><tr><td>T100</td><td>{A,B,D }</td></tr><tr><td>T200</td><td>{A,B,C }</td></tr><tr><td>T300</td><td>{B,C,D,E }</td></tr></table>	TID	Items_Bought	T100	{A,B,D }	T200	{A,B,C }	T300	{B,C,D,E }	4	3	7						
TID	Items_Bought																		
T100	{A,B,D }																		
T200	{A,B,C }																		
T300	{B,C,D,E }																		

		<table><tr><td>T400</td><td>{A,C,E }</td></tr><tr><td>T500</td><td>{A,B,C,E }</td></tr><tr><td>T600</td><td>{A,B,D,E}</td></tr></table>	T400	{A,C,E }	T500	{A,B,C,E }	T600	{A,B,D,E}																																														
T400	{A,C,E }																																																					
T500	{A,B,C,E }																																																					
T600	{A,B,D,E}																																																					
	b).	Explain confidence-based pruning of association rules stating the relevant theorem.	4	2	3																																																	
		UNIT-5																																																				
10.	a).	Apply K-means clustering to form 2 clusters from the following data points: A(5,3) B(-3,4) C(2,4) D(6,5), E(4,3) Consider Euclidean distance. Consider C and D as initial cluster centers. Perform 2 iterations.	5	3	7																																																	
	b).	Explain briefly about well-separated and center-based clusters.	5	3	3																																																	
		OR																																																				
11.	a).	Apply the approach of single link and perform agglomerative clustering of the points whose pair-wise distance is as follows:<table><tr><td></td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td><td>P6</td></tr><tr><td>P1</td><td>0.00</td><td>0.27</td><td>0.25</td><td>0.35</td><td>0.33</td><td>0.26</td></tr><tr><td>P2</td><td>0.27</td><td>0.00</td><td>0.23</td><td>0.25</td><td>0.22</td><td>0.29</td></tr><tr><td>P3</td><td>0.25</td><td>0.23</td><td>0.00</td><td>0.23</td><td>0.28</td><td>0.11</td></tr><tr><td>P4</td><td>0.35</td><td>0.25</td><td>0.23</td><td>0.00</td><td>0.33</td><td>0.29</td></tr><tr><td>P5</td><td>0.33</td><td>0.22</td><td>0.28</td><td>0.33</td><td>0.00</td><td>0.40</td></tr><tr><td>P6</td><td>0.26</td><td>0.29</td><td>0.11</td><td>0.29</td><td>0.40</td><td>0.00</td></tr></table> Draw dendrogram also.		P1	P2	P3	P4	P5	P6	P1	0.00	0.27	0.25	0.35	0.33	0.26	P2	0.27	0.00	0.23	0.25	0.22	0.29	P3	0.25	0.23	0.00	0.23	0.28	0.11	P4	0.35	0.25	0.23	0.00	0.33	0.29	P5	0.33	0.22	0.28	0.33	0.00	0.40	P6	0.26	0.29	0.11	0.29	0.40	0.00	5	3	7
	P1	P2	P3	P4	P5	P6																																																
P1	0.00	0.27	0.25	0.35	0.33	0.26																																																
P2	0.27	0.00	0.23	0.25	0.22	0.29																																																
P3	0.25	0.23	0.00	0.23	0.28	0.11																																																
P4	0.35	0.25	0.23	0.00	0.33	0.29																																																
P5	0.33	0.22	0.28	0.33	0.00	0.40																																																
P6	0.26	0.29	0.11	0.29	0.40	0.00																																																
	b).	Explain core, object, and noise points in DBSCAN approach with a suitable example.	5	2	3																																																	

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
CLOUD COMPUTING					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain Cloud computing briefly.	1	1	2
	b).	Defining a cloud.	1	1	2
	c).	Explain about parallel and distributed computing.	2	1	2
	d).	Describe elements of distributed computing?	2	1	2
	e).	Define Virtualization.	3	1	2
	f).	Describe about Docker Swarm and Kubernetes.	3	1	2
	g).	Explain Economics of the cloud.	4	1	2
	h).	Define cloud computing security.	4	1	2
	i).	Explain Serverless computing.	5	1	2
	j).	Define edge and fog computing.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain cloud computing reference model.	1	2	5
	b).	Describe types of services (IaaS, PaaS, SaaS).	1	2	5
		OR			
3.	a).	Explain cloud deployment models (public, private, hybrid).	1	2	5
	b).	Explain cloud service providers (Amazon Web Services, Microsoft Azure, Google App-Engine.	1	2	5
		UNIT-2			
4.	a).	Explain parallel and distributed computing.	2	2	5

	b).	Demonstrate hardware architectures for parallel computing (SISD, SIMD, MISD, MIMD).	2	2	5
		OR			
5.	a).	Discuss the elements of distributed computing, Inter-process communication.	2	2	5
	b).	Discuss about Remote Procedure Calls (RPC), service-oriented architecture (SOA).	2	2	5
		UNIT-3			
6.	a).	Explain the Characteristics of virtualized environments.	3	2	5
	b).	Discuss about taxonomy of virtualization techniques.	3	2	5
		OR			
7.	a).	Explain the technology examples (XEN, VMware).	3	2	5
	b).	Explain about the building blocks of containers, container platforms (LXC, Docker).	3	2	5
		UNIT-4			
8.	a).	Discuss the Economics of the cloud, cloud interoperability and standards.	4	2	5
	b).	Explain about cloud computing security, fundamentals of computer security.	4	2	5
		OR			
9.	a).	Explain about cloud security architecture, cloud shared responsibility model.	4	2	5
	b).	Discuss about security in cloud deployment models.	4	2	5
		UNIT-5			
10.	a).	Discuss about Serverless computing, Function-as-a-Service, serverless computing architecture.	5	2	5
	b).	Explain Public cloud (e.g. AWS Lambda) and open source (e.g. OpenFaaS) serverless platforms.	5	2	5
		OR			
11.	a).	Explain the cloud centric IoT - IoT architecture and cloud layers.	5	2	5
	b).	Explain infrastructure-as-code, Quantum-as-a-Service (QaaS).	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3202					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
CRYPTOGRAPHY & NETWORK SECURITY					
For IT					
Time:3 Hrs			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the key principles of security?	1	1	2
	b).	Distinguish between passive attacks and active attacks?	1	1	2
	c).	Compare symmetric and asymmetric key cryptography.	2	1	2
	d).	What is the need for Diffie-Hellman key exchange?	2	1	2
	e).	What is a hash function?	3	1	2
	f).	Differentiate between Message digest and MAC	3	1	2
	g).	Write the four SSL Protocols.	4	1	2
	h).	What are the services provided by IPSec?	4	1	2
	i).	List out the design goals of firewalls.	5	1	2
	j).	What is VPN	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain Euler's Theorem by solving $3^{202} \bmod 13$.	1	2	5
	b).	Find the value of X such that $2X \equiv 6 \pmod{14}$ $3X \equiv 9 \pmod{15}$ $5X \equiv 20 \pmod{60}$ using Chinese Remainder Theorem	1	2	5
		OR			
3.	a).	Explain about Security approaches	1	2	5
	b).	Explain about types of attacks	1	2	5

		UNIT-2			
4.	a).	What are the substitution techniques used in cryptography?	2	2	5
	b).	Explain the procedure to encrypt a given data using Data Encryption Standard Algorithm.	2	2	5
		OR			
5.	a).	Explain about RC4 algorithm.	2	2	5
	b).	Explain Transposition techniques in cryptography.	2	2	5
		UNIT-3			
6.	a).	Justify Man in Middle attack with an example.	3	2	5
	b).	Perform encryption and decryption using RSA algorithm for the following data N=187, E=17. PT(Plain Text)=10	3	2	5
		OR			
7.	a).	Explain MD5 algorithm.	3	2	5
	b).	Explain HMAC algorithm.	3	2	5
		UNIT-4			
8.	a).	Explain creation of digital certificates.	4	2	5
	b).	Explain about Authentication tokens.	4	2	5
		OR			
9.	a).	Explain about Kerberos as authentication protocol	4	2	5
	b).	Write a Short notes on Authentication mechanisms	4	2	5
		UNIT-5			
10.	a).	Explain SSL protocol	5	2	5
	b).	Explain Email security protocols	5	2	5
		OR			
11.	a).	Explain about Operation of IPSec protocol	5	2	5
	b).	Write short notes on fire walls and configuration of firewalls	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23IT3203					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MACHINE LEARNING					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Compare AL,ML,DL	1	2	2
	b).	Define Bias and Variance	1	2	2
	c).	List the types of regression	2	1	2
	d).	Define Binary classification	2	1	2
	e).	List the type of learners	1	1	2
	f).	Outline the limitations of KNN	3	2	2
	g).	Compare SVM with SVR	3	2	2
	h).	Outline the measures of similarity and dis-similarity in cluster analysis	4	2	2
	i).	Explain about partitioning methods	5	2	2
	j).	Outline the importance of perceptron	5	2	2
					5 x 10 = 50 Marks
		UNIT-1	CO	KL	M
2.	a).	Explain about different types of machine learning	1	2	5
	b).	Explain about feature selection and extraction	1	2	5
		OR			
3.	a).	Explain any 7 challenges of Machine learning	1	2	5
	b).	Contrast between Overfitting and Underfitting	1	2	5
		UNIT-2			
4.		Differentiate between Linear and Non linear regression	2	2	10
		OR			
5.	a).	Explain about regularization and its types	2	2	5
	b).	Summarize the importance of logistic regression	2	2	5
		UNIT-3			
6.		Outline Binary classifier,Multi class classifier,Multi label classification and imbalanced classification	2	2	10

		OR																								
7.	a).	Illustrate about distance metrics	3	2	5																					
	b).	Explain the key assumptions of Naive Bayes and describe how it calculates the probability of a class given the feature values.	4	2	5																					
		UNIT-4																								
8.		Summarize Ensembling learning technics	4	2	10																					
		OR																								
9.		Analyze how PCA reduces dimensionality while retaining the maximum variance in the data.	3	3	10																					
		UNIT-5																								
10.	a).	Develop K Means algorithm for the following data with 2 clusters	3	3	5																					
		<table><tr><td>Entity</td><td>X1</td><td>X2</td></tr><tr><td>A</td><td>1.0</td><td>2.0</td></tr><tr><td>B</td><td>1.5</td><td>1.8</td></tr><tr><td>C</td><td>1.2</td><td>1.9</td></tr><tr><td>D</td><td>5.0</td><td>8.0</td></tr><tr><td>E</td><td>6.0</td><td>9.0</td></tr><tr><td>F</td><td>5.5</td><td>7.5</td></tr></table>				Entity	X1	X2	A	1.0	2.0	B	1.5	1.8	C	1.2	1.9	D	5.0	8.0	E	6.0	9.0	F	5.5	7.5
		Entity				X1	X2																			
		A				1.0	2.0																			
		B				1.5	1.8																			
		C				1.2	1.9																			
		D				5.0	8.0																			
		E				6.0	9.0																			
F	5.5	7.5																								
	b).	Explain reinforcement learning and its types	4	2	5																					
		OR																								
11.	a).	Explain the basic structure of an Artificial Neural Network (ANN) .	4	2	5																					
	b).	Analyze how the Backpropagation algorithm updates the weights to minimize the error during the training process.	4	3	5																					

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
SOFTWARE TESTING METHODOLOGIES					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate between effective and exhaustive software testing.	1	2	2
	b).	Explain any two software testing methodologies.	1	2	2
	c).	Identify Boundary Value Analysis in black-box testing?	2	1	2
	d).	Define Logic Coverage criterion in white-box testing?	2	1	2
	e).	List some advantages of using Inspections in software testing?	3	1	2
	f).	Explain the purpose of Integration Testing.	3	2	2
	g).	Define test suite prioritization, and why is it important?	4	1	2
	h).	Describe any two debugging techniques.	4	2	2
	i).	List some key guidelines for effective automated testing.	5	1	2
	j).	Discuss the major challenges in testing web-based software?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the primary goals of software testing, and how does psychology play a role in the testing process?	1	2	5
	b).	Explain different validation techniques with examples.	1	2	5
		OR			
3.	a).	Discuss different software testing methodologies and their advantages and disadvantages.	1	2	5
	b).	Explain various verification activities performed during the software development life cycle.	1	2	5
		UNIT-2			
4.		Explain the concept of Black Box Testing and discuss its importance in software testing.	2	2	10
		OR			
5.		Explain Data Flow Testing and its significance in detecting software anomalies.	2	2	10

		UNIT-3			
6.	a).	Define regression testing. Explain different types of regression testing.	3	2	6
	b).	Discuss the objects of regression testing	3	2	4
		OR			
7.	a).	Define functional testing? How does it ensure that software meets user requirements?	3	2	5
	b).	Define unit testing? Explain its role in software quality assurance	3	2	5
		UNIT-4			
8.		Discuss the importance of minimizing a test suite. What are the key benefits of test suite minimization?	4	2	10
		OR			
9.		Explain software quality metrics? Discuss different types of metrics used to measure software quality.	4	2	10
		UNIT-5			
10.	a).	Explain the different categories of testing tools with examples.	5	2	6
	b).	Differentiate between testing object-oriented software and traditional procedural software?	5	2	4
		OR			
11.	a).	Summarize unique challenges faced in mobile application testing, and how can they be addressed?	5	2	5
	b).	Distinguish between web-based and web-enabled applications?	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3205					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
AUGMENTED REALITY & VIRTUAL REALITY					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe augmented reality?	1	2	2
	b).	Discuss about any two Mobile Sensors Applications	1	2	2
	c).	Compare Output Modalities and Input Modalities	2	2	2
	d).	Explain briefly Scene Graphs?	2	2	2
	e).	Explain about Basic Behaviour of Light	3	2	2
	f).	Describe about Viewing Transformations	3	2	2
	g).	Elaborate any two Types of Eye Movements	3	2	2
	h).	Brief about colour Visual Rendering?	4	2	2
	i).	Explain a Virtual World?	4	2	2
	j).	Describe the Auditory Rendering	4	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain augmented reality with examples.	1	2	10
		OR			
3.	a).	Describe Multimodal Displays with an example	1	2	5
	b).	Discuss the Characteristics of Tracking Technology	1	2	5
		UNIT-2			
4.	a).	Illustrate Virtual User Interfaces on Real Surfaces.	2	2	5
	b).	Explain about Hardware and software Requirements of AR	2	2	5
		OR			
5.	a).	Outline the applications of augmented reality in detail	2	2	5
	b).	Outline the Dataflow Architecture in AR with an example	2	2	5
		UNIT-3			
6.	a).	Explain Human Physiology and Perception in VR.	3	2	10
		OR			

7.	a).	Discuss about different Types of Aberrations in VR	3	2	5
	b).	Detail about axis-angle representations of rotation with an example.	3	2	5
		UNIT-4			
8.	a).	Explain the Key Aspects of Visual Perception in VR	3	2	5
	b).	Discuss the Photoreceptors with a diagram.	3	2	5
		OR			
9.	a).	Summarize Latency and Frame Rates in VR	4	2	5
	b).	Explain the positive and negative Distortion	4	2	5
		UNIT-5			
10.	a).	Explain the Audio of physiology of hearing with a diagram .	4	2	10
		OR			
11.	a).	Describe the Key Aspects of Social Interaction with an example	4	2	5
	b).	Explain the applications of virtual reality in detail	4	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks



III B.Tech. II Semester MODEL QUESTION PAPER

DevOps

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Detail the DevOps lifecycle with a simple example.	1	2	2
	b).	Describe the key principles of DevOps?	1	2	2
	c).	Discuss Why is version control important in software development?	2	2	2
	d).	Explain how unit testing supports the CI/CD processes?	2	2	2
	e).	Describe the role of Jenkins in build automation?	3	2	2
	f).	Elaborate the purpose of scheduling builds in Jenkins?	3	2	2
	g).	Explain how DockerFile help in container creation?	4	2	2
	h).	Discuss the key features of Selenium for automated testing?	4	2	2
	i).	Discuss the purpose of Ansible in DevOps automation?	5	2	2
	j).	Compare Puppet and Chef in configuration management?	5	2	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M
2.	a).	Describe the DevOps architecture and its main components.	1	2	5
	b).	List common DevOps tools and explain their use across different stages of the DevOps lifecycle.	1	2	5
		OR			
3.	a).	Discuss about build automation and how does it benefit the development process?	1	2	5
	b).	Compare Scrum and Kanban models used in DevOps project management	1	2	5
		UNIT-2			
4.	a).	Discuss about version control? Explain its importance in modern software development.	2	2	5
	b).	Elaborate the basic Git workflow including commit, push, and pull operations.	2	2	5
		OR			
5.	a).	Justify the role of unit testing and its impact on software quality.	2	2	5

	b).	Describe how SonarQube helps in maintaining code quality?	2	2	5
		UNIT-3			
6.	a).	Apply the concept of Jenkins pipelines to automate the build and test process of a Java project. Explain the steps.	3	3	5
	b).	Create a Jenkins freestyle project that includes Git integration and test execution using JUnit.	3	3	5
		OR			
7.	a).	Write a simple pipeline using Jenkinsfile that includes stages: build, test, and deploy.	3	3	5
	b).	Apply the concept of user roles in Jenkins to set up user management and access control in a CI pipeline.	3	3	5
		UNIT-4			
8.	a).	Apply the concept of Continuous Delivery to automate deployment of a web application after successful testing.	4	3	5
	b).	Illustrate the steps to push a Docker image to Docker Hub after successful build and test.	4	3	5
		OR			
9.	a).	Apply Docker commands to manage containers, such as creating, listing, and removing them. Provide an example.	4	3	5
	b).	Write a Selenium script in Java to test login functionality of a sample website.	4	3	5
		UNIT-5			
10.	a).	Demonstrate writing a basic Ansible playbook to install and start Apache web server on a remote server.	5	3	5
	b).	Demonstrate the use of ConfigMaps in Kubernetes to manage configuration data separately from code.	5	3	5
		OR			
11.	a).	Demonstrate the deployment of a simple application on OpenShift using DeploymentConfig (DC) and ConfigMaps	5	3	5
	b).	Apply Puppet or Chef to automate configuration of a multi-node system. Explain with an example.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

III B.Tech. II Semester MODEL QUESTION PAPER

GENERATIVE AI

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define generative modeling in artificial intelligence.	1	1	2
	b).	List any two types of generative models.	1	1	2
	c).	What is a language model?	2	1	2
	d).	Name any two transformer-based text generation models.	2	1	2
	e).	Expand the acronym GAN.	3	1	2
	f).	Mention any two applications of image generation models.	3	1	2
	g).	What is CycleGAN used for?	4	1	2
	h).	Name any one generative model for music and one for painting.	4	1	2
	i).	Name any two open-source frameworks used in generative AI.	5	1	2
	j).	What is LangChain mainly used for in generative AI?	5	1	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M
2.	a).	Explain the difference between generative and discriminative modeling in AI with examples.	1	2	5
	b).	Discuss the future directions and ethical challenges associated with generative AI.	1	2	5
		OR			
3.	a).	Describe various types of generative models such as GANs, VAEs, and autoregressive models.	1	2	5
	b).	Describe the importance of generative modeling in AI and machine learning.	1	2	5
		UNIT-2			
4.	a).	Describe the architecture and working of transformer-based language models like GPT.	2	2	5
	b).	Apply prompt engineering strategies to improve text generation outputs.	2	3	5
		OR			
5.	a).	Explain the key building blocks of language models, including encoder-	2	2	5

		decoder and attention mechanisms.			
	b).	Explain Retrieval-Augmented Generation (RAG) and its application in multimodal LLMs.	2	3	5
		UNIT-3			
6.	a).	Describe the working of GANs and the adversarial training process.	3	2	5
	b).	Apply stable diffusion and transformer-based techniques like CLIP and DALL·E for image generation.	3	3	5
		OR			
7.	a).	Explain the role of encoder-decoder architectures in VAEs and diffusion models.	3	2	5
	b).	Illustrate how visual transformers (ViT) enhance the quality of generated images.	3	3	5
		UNIT-4			
8.	a).	Explain the concept of Neural Style Transfer and its applications in digital painting.	4	2	5
	b).	Apply MuseGAN or music-generating RNNs to synthesize new music patterns.	4	3	5
		OR			
9.	a).	Describe the architecture and use of CycleGANs for style and domain transformation.	4	2	5
	b).	Discuss the use of generative models in training autonomous agents using reinforcement learning.	4	3	5
		UNIT-5			
10.	a).	Describe the training and fine-tuning process of generative models using pre-trained architectures	5	2	5
	b).	Apply open-source frameworks like Hugging Face any generative AI project	5	3	5
		OR			
11.	a).	Explain the use of transfer learning in accelerating the development of generative models.	5	2	5
	b).	Illustrate the deployment pipeline of a generative vision model using Hugging Face or Google Copilot.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3209					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
SOFTWARE PROJECT MANAGEMENT					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define software economics in the context of software development	1	1	2
	b).	List two ways to improve team effectiveness in software projects.	1	1	2
	c).	Give one example each of a management artifact and an engineering artifact.	2	1	2
	d).	Outline the objective of the inception phase in the software life cycle?	2	2	2
	e).	Explain the purpose of software process workflow.	3	2	2
	f).	Relate pragmatic planning importance in iterative software development?	3	2	2
	g).	Define Line of business(LOB)	4	1	2
	h).	List two key responsibilities of a Project Manager.	4	1	2
	i).	Define Scrum in the context of Agile.	5	1	2
	j).	List any 2 technological aspects important for DevOps adoption	5	1	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Outline the activities of software project management with an example.	1	2	5
	b).	Explain the limitations of the Waterfall model and discuss how it affects the performance of conventional software management.	1	2	5
		OR			
3.	a).	Compare the principles of conventional software engineering with those of modern software management	1	2	5
	b).	Identify the factors influencing software economics and explain how pragmatic cost estimation methods improve project planning.	1	3	5
		UNIT-2			
4.		Illustrate software Life cycle phases.	2	2	10
		OR			
5.		Differentiate between management artifacts, engineering artifacts, and programmatic artifacts with suitable examples for each.	2	2	10

		UNIT-3			
6.	a).	Compare between the management perspective and the technical perspective of model-based software architecture.	3	2	6
	b).	Demonstrate about iteration workflow.	3	2	4
		OR			
7.		Identify the different types of checkpoints in the software development process.	3	3	10
		UNIT-4			
8.	a).	Identify the key building blocks of process automation in software development.	4	3	5
	b).	Explain the components of the project environment.	4	2	5
		OR			
9.		Outline the seven-core metrics used in software project control.	4	2	10
		UNIT-5			
10.		Describe the Scrum framework and explain how organizations can adopt Scrum using different patterns.	5	2	10
		OR			
11.		Utilize the DevOps ecosystem and explain how its different elements interact to support agile software delivery.	5	3	10

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3210					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MOBILE ADHOC NETWORKS					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Compare a MANET with a cellular network.	1	2	2
	b).	List any four applications of Ad Hoc Networks.	1	2	2
	c).	Explain the reasons for poor performance of conventional TCP in Ad Hoc networks.	2	2	2
	d).	Discuss the factors that make routing more difficult in Ad Hoc networks compared to wired networks.	2	2	2
	e).	Explain the importance of key management in wireless security.	3	2	2
	f).	Discuss about secure routing and why is it needed in MANETs?	3	2	2
	g).	Explain the importance of low energy usage in sensor networks?	4	2	2
	h).	Describe the process of data collection and transmission in sensor networks.	4	2	2
	i).	Describe the role of TinyOS in a sensor network.	5	2	2
	j).	Explain the role of NS-2 in simulating sensor networks.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the characteristics of Ad Hoc Networks with suitable examples.	1	2	5
	b).	Explain issues and challenges faced by MANETs.	1	2	5
		OR			
3.	a).	Discuss the major issues to consider while designing MAC protocols for Ad Hoc networks?	1	2	5
	b).	Describe the classification of MAC protocols used in Ad Hoc Wireless Networks.	1	2	5
		UNIT-2			
4.	a).	Demonstrate the application of different types of transport layer protocols in mobile Ad Hoc networks.	2	3	5
	b).	Demonstrate the operational differences between topology-based and position-based routing protocols.	2	3	5

		OR			
5.	a).	Apply the classification of TCP protocols in MANETs to explain how each type helps to solve problems like mobility or packet loss.	2	3	5
	b).	Use suitable strategies to solve the challenges faced by TCP in Ad Hoc wireless networks.	2	3	5
		UNIT-3			
6.	a).	Explain the importance of security in Ad Hoc Wireless Networks.	3	2	5
	b).	Discuss the main issues and challenges involved in providing security for Ad Hoc Wireless Networks.	3	2	5
		OR			
7.	a).	Explain the different types of network security attacks in MANETs.	3	2	5
	b).	Describe the role of Intrusion Detection Systems in securing Ad Hoc Wireless Networks.	3	2	5
		UNIT-4			
8.	a).	Demonstrate the use of the Mica Mote to enhance the efficiency of a wireless sensor network.	4	3	5
	b).	Implement a data retrieval strategy for a WSN with limited power and bandwidth resources.	4	3	5
		OR			
9.	a).	Demonstrate the impact of design issues such as power, size, and cost on sensor network deployment.	4	3	5
	b).	Apply dynamic routing techniques in WSNs to handle frequent topology changes.	4	3	5
		UNIT-5			
10.	a).	Summarize the key management techniques used in WSNs.	5	2	5
	b).	Discuss the ways in which LA-TinyOS improves upon standard TinyOS.	5	2	5
		OR			
11.	a).	Explain the role of secure data aggregation in protecting data in WSNs.	5	2	5
	b).	Discuss the working of TOSSIM as a simulator for WSN applications.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

III B.Tech. II Semester MODEL QUESTION PAPER

NATURAL LANGUAGE PROCESSING

For IT

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define tokenization in NLP.	1	2	2
	b).	What is the role of regular expressions in text processing?	1	2	2
	c).	Define Part-of-Speech (PoS) tagging.	2	2	2
	d).	What is the significance of word classes in NLP?	2	2	2
	e).	What is a Treebank?	3	2	2
	f).	Define shallow parsing.	3	2	2
	g).	What are semantic attachments in syntax trees?	4	2	2
	h).	Define word sense in the context of NLP.	4	2	2
	i).	What is the purpose of Porter Stemmer in NLP?	5	2	2
	j).	Define Coreference Resolution.	5	2	2

5 x 10 = 50 Marks

		UNIT-1	CO	KL	M
2.	a).	Explain the challenges in Natural Language Processing (NLP).	1	2	5
	b).	Describe the concept of Minimum Edit Distance and explain how it is calculated.	1	2	5
		(OR)			
3.	a).	Describe statistical language modeling and compare it with grammar-based language modeling.	1	2	5
	b).	Explain the use of Finite-State Automata in NLP with relevant examples.	1	2	5
		UNIT-2			
4.	a).	Apply the Hidden Markov Model to demonstrate the functioning of a PoS tagger using a sample sentence.	2	3	5
	b).	Implement and compare different smoothing techniques in N-gram models.	2	3	5
		(OR)			
5.	a).	Analyze and contrast Rule-based, Stochastic, and Transformation-based	2	3	5

		PoS tagging methods through an example.			
	b).	Apply interpolation and backoff techniques to a given N-gram language modeling task.	2	3	5
		UNIT-3			
6.	a).	What is a Context-Free Grammar (CFG)? Explain its use in syntactic parsing.	3	2	5
	b).	Explain the purpose of probabilistic CFGs in syntactic parsing with the help of an example.	3	2	5
		(OR)			
7.	a).	Describe the concept of dynamic programming in syntactic parsing.	3	2	5
	b).	Discuss the role of feature structures and unification in NLP.	3	2	5
		UNIT-4			
8.	a).	Analyze and compare supervised and dictionary-based Word Sense Disambiguation (WSD) techniques.	4	3	5
	b).	Apply the concepts of thematic roles and selectional restrictions to identify semantic roles in a sentence.	4	3	5
		(OR)			
9.	a).	Implement Syntax-Driven Semantic Analysis using an example sentence.	4	3	5
	b).	Measure and compare word similarity using thesaurus-based and distributional methods.	4	3	5
		UNIT-5			
10.	a).	What is the Anaphora Resolution? Describe Hobbs and Centering Algorithms.	5	2	5
	b).	Describe the role of WordNet and FrameNet in NLP.	5	2	5
		(OR)			
11.	a).	What is discourse segmentation? Explain reference phenomena and coherence with examples.	5	2	5
	b).	Write short notes on: (i) Penn Treebank (ii) Brill's Tagger (iii) British National Corpus	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks

Course Code: B23IT3212					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
DISTRIBUTED OPERATING SYSTEMS					
For IT					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define distributed operating systems.	1	1	2
	b).	Explain Remote Procedure Call (RPC) ?	1	2	2
	c).	Differentiate between logical clocks and physical clocks ?	2	2	2
	d).	Explain deadlocks in distributed operating systems ?	2	2	2
	e).	How to implement threads package in distributed operating systems ?	3	1	2
	f).	Compare processor allocation models ?	3	2	2
	g).	Explain caching and replication in distributed file systems ?	4	2	2
	h).	List out the semantics of file sharing in distributed file systems ?	4	1	2
	i).	What is Distributed Shared Memory (DSM) ?	5	1	2
	j).	Explain about bus, ring in distributed shared memory ?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Discuss the semantics of RPC in the presence of failure.	1	2	5
	b).	Illustrate hardware architectures with the help of diagrams.	1	2	5
		OR			
3.	a).	Explain various choices of addressing, blocking in Client-Server communication ?	1	2	5
	b).	Explain various types of operating system and analyze its design issues?	1	2	5
		UNIT-2			
4.	a).	Design the Bully algorithm with the help of a diagram.	2	3	5
	b).	Implement Mutual Exclusion-Centralized Algorithm.	2	3	5
		OR			
5.	a).	Define Clock Synchronization. Differentiate logical clocks and physical clocks ?	2	2	5
	b).	Explain the system model for distributed mutual exclusion algorithm and identify the requirements ?	2	2	5

		UNIT-3			
6.	a).	Identify design issues for threads package and implement threads package with an example ?	3	2	5
	b).	Differentiate between Processor pool model and Hybrid model ?	3	2	5
		OR			
7.	a).	Explain various process allocation models ?	3	2	5
	b).	Explain the workstation model to organize multiple processors in a distributed system ?	3	2	5
		UNIT-4			
8.		Identify the semantics of file sharing and explain file usage, system structure ?	4	2	10
		OR			
9.	a).	Explain the different ways of replication ?	4	2	5
	b).	Explain Caching in Distributed File System implementation ?	4	2	5
		UNIT-5			
10.		Explain about Consistency Models ?	5	2	10
		OR			
11.	a).	Explain the working of Bus based multiprocessors ?	5	2	5
	b).	Explain Object based Distributed Shared Memory (DSM) ?	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A, B splits or as a single Question for 10 marks